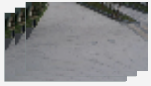
VIDEO



every 16th frame



current frame f_t

FIXED TEXT

59 phrases
11 + 48

fixed before scoring;
same for every camera

no caption, no sentence
per camera

❄ = frozen

M1 · VISUAL SCORING

❄ **PE-Core image encoder**

E_I · PE-Core-L/14 · 336 px

$$v_t = E_I(f_t) \in \mathbb{R}^{1024}$$

each frame becomes 1024 numbers of unit length;
the direction encodes what it contains

M3 · NORMAL AND ABNORMAL TEXT

C · calm normal
"a person standing still"

B · action bank
"two people punching each other"

P^+ normal
3 general + 8 calm

P^- abnormal
12 types × 4 actions

❄ **PE-Core text encoder E_T**

FRAME SCORING

e^+ e^-

abnormal minus normal

$$m_t = \langle v_t, e^- \rangle - \langle v_t, e^+ \rangle$$

$$\rightarrow\ s_t = \sigma(\lambda\, m_t)$$

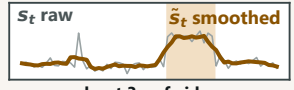
$e^{\pm}$: each side's mean, renormalised
λ = 100, fixed

s_t

M2 · SMOOTHING

centred moving average
window w = 5 frames

$$\tilde{s}_t = \frac{1}{w} \sum_{i=t-2}^{t+2} s_i$$

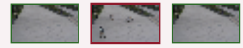


about 3 s of video:
one spike cannot raise an alarm

DETECTION

flag if $\tilde{s}_t \geq \tau$

τ: the alarm threshold



a score for every frame

not used

scene sentence
captioning model
target-site data
gradient updates

run-time cost

one image pass per
sampled frame;
text encoded once

New camera = no new data · the same 59 phrases · no labels · no gradients · every model frozen

Evaluation: frame-level AUROC (AP on XD-Violence); ShanghaiTech uses every 2nd frame, w = 31 and per-clip normalisation, its benchmark protocol